

Using Cough Sounds for the Recognition of Multiclass Respiratory Diseases with Artificial Intelligence: A Review

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ABSTRACT The multi-class identification of respiratory diseases based on cough sounds plays a critical role in the diagnosis of respiratory conditions. Compared to traditional auscultation methods, multi-class techniques integrate various deep-level disease features of cough sounds, enabling a more comprehensive capture of cough characteristics associated with specific diseases, thereby effectively reducing the risk of misdiagnosis. Furthermore, multiclass classification enhances the system's ability to differentiate among multiple respiratory conditions, significantly improving diagnostic accuracy and robustness. This study provides an in-depth examination of artificial intelligence-driven methodologies for multi-class respiratory disease recognition. We reviewed 19 articles from a 5-year open-access literature database to summarize the current research on diagnosing voice-affecting disorders through AI techniques using cough sound samples for multi-class classification. The study also covers AI workflows for multi-class cough recognition, including datasets, feature extraction techniques, models, and evaluation metrics. Additionally, recent advancements in AI technologies, cough monitoring applications, and wearable devices are discussed, along with the challenges and future research directions aimed at developing robust and universally applicable solutions for multi-class diagnosis.

INDEX TERMS Cough sounds, respiratory diseases, multi-class, deep learning

I. INTRODUCTION

A. BACKGROUND AND MOTIVATION

Respiratory diseases represent one of the most significant global health challenges of the 21st century, affecting

millions across socioeconomic boundaries and healthcare systems worldwide [1]. According to the World Health Organization, cardiovascular and respiratory conditions emerged as the predominant causes of mortality worldwide in 2021. Notably, COVID-19 ranked as the second leading cause of global mortality, underscoring the critical impact

of respiratory diseases on public health [2]. This escalating health burden necessitates sophisticated diagnostic frameworks capable of early pathology detection and precise disease characterization.

Early studies primarily focused on binary classification tasks. These binary approaches, however, fail to address three critical challenges: (1) The issue of generalizability in disease diagnostic systems for clinical applications. [3], (2) insufficient capability to capture subtle pathological variations [4], and (3) challenges in distinguishing between multiple respiratory conditions simultaneously [5].

B. TECHNICAL EVOLUTION

The field of respiratory disease diagnostics has witnessed significant methodological evolution. Initial research focused on single-disease diagnosis, such as tuberculosis [6], [7] and asthma [8], gradually expanding to multi-disease frameworks encompassing conditions like asthma, COPD, and COVID-19 [9]. This evolution has been particularly evident in COVID-19 research, where approaches have progressed from single-modality methods utilizing speech or cough sounds [10], [11], [12], [13], [14] to sophisticated multi-modal frameworks integrating various data types [15], [16], [17], [18].

Technical innovations have emerged along multiple dimensions: enhanced model interpretability [19], [20], [21], advanced algorithmic approaches [22], [23], and resource-efficient implementations [24], [25]. These developments have facilitated the creation of comprehensive cough sound analysis systems capable of analyzing multiple diseases [26], [27], [28] and supporting real-time clinical applications.

C. CLINICAL IMPACT

Recent advancements in multi-class respiratory disease classification using cough sounds have demonstrated remarkable clinical potential [29]. Moving beyond traditional binary approaches, these systems have achieved unprecedented diagnostic accuracy in distinguishing multiple respiratory conditions simultaneously: 99.39% accuracy in four-class classification [5], 99.01% accuracy across nine distinct respiratory conditions [30], and F1-scores of 0.8631 in three-class respiratory disease detection [31]. The superiority of multi-class approaches is evidenced through several key innovations: channel attention mechanisms for feature discrimination [32], multi-modal integration frameworks for comprehensive disease characterization [30], and dynamic feature selection for improved class separation [5]. Furthermore, these multi-class systems have demonstrated robust performance in real-world clinical settings [33], effectively addressing the complex challenge of simultaneous disease identification. These advances highlight the transformative potential of

multi-class classification approaches in enhancing diagnostic precision and efficiency in respiratory healthcare [34].

D. RESEARCH GAPS AND CHALLENGES

The field of multi-class respiratory disease classification using cough sounds faces several research gaps [29]. Firstly, systematic evaluations of multi-class classification approaches are lacking [35], with limited standardization of performance metrics [36] and insufficient exploration of model interpretability [37]. Secondly, current methods struggle to address complex real-world scenarios [38], including coexisting respiratory conditions [1], the influence of environmental noise, and patient-specific variability [39]. Lastly, validation efforts are inadequate, with a shortage of large-scale clinical trials, cross-population and cross-regional studies [40], and longitudinal investigations to assess the long-term applicability of these systems. These gaps highlight the pressing need for targeted research to address these limitations [41].

E. RESEARCH SCOPE AND SIGNIFICANCE

This systematic review focuses on multi-class respiratory disease classification through AI-based cough sound analysis, addressing three objectives:

1. Analyze the transition from binary to multi-class respiratory classification, examining AI frameworks for simultaneous differentiation of multiple respiratory conditions.
2. Establish evaluation frameworks for multi-class respiratory classification, addressing challenges of class imbalance and overlapping acoustic features.
3. Identify opportunities in multi-class classification architectures and feature extraction methods that enhance multi-disease discrimination.

The significance lies in providing the first systematic analysis of multi-class respiratory classification approaches, establishing methodological foundations for automated multi-disease diagnosis.

F. METHODOLOGICAL FRAMEWORK

To address these methodological challenges, Figure 1 delineates a systematic framework for developing AI-enabled acoustic diagnostic systems. This comprehensive pipeline encompasses four integral stages: (1) Acoustic Data Acquisition and Preprocessing: Establishing protocols to ensure signal standardization. (2) Advanced Augmentation and Feature Extraction: Facilitating robust representation learning. (3) Sophisticated Model Architecture Development: Enhancing classification performance through optimization. (4) Rigorous Evaluation Protocols: Ensuring reliable clinical deployment. This end-

to-end pipeline ensures robust system development from initial data processing to final clinical integration.

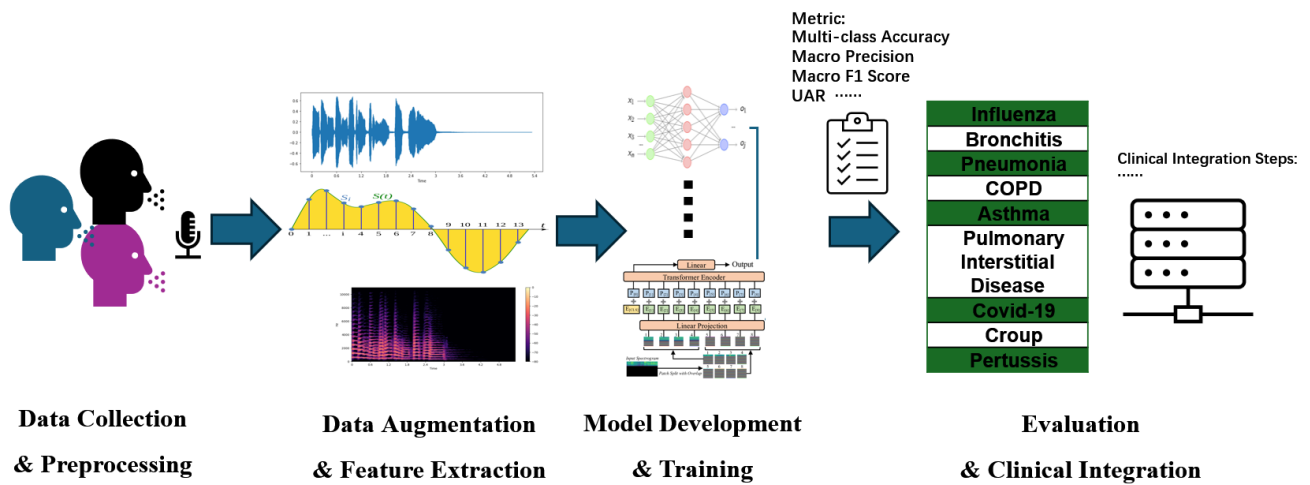


FIGURE 1. AI Workflows for Multi-Class Cough Recognition.

G. REVIEW METHODOLOGY

This comprehensive systematic review was conducted in accordance with PRISMA guidelines. The literature search including PubMed, ScienceDirect, Springer, and Google Scholar, focusing on peer-reviewed English-language publications from May 2019 to Dec 2024. The systematic

search protocol initially identified 149 relevant studies. Following rigorous screening procedures, including duplicate removal and comprehensive evaluation of titles and abstracts against predefined inclusion criteria, 19 articles were selected for detailed qualitative analysis, ensuring comprehensive coverage of significant methodological advances in the field.

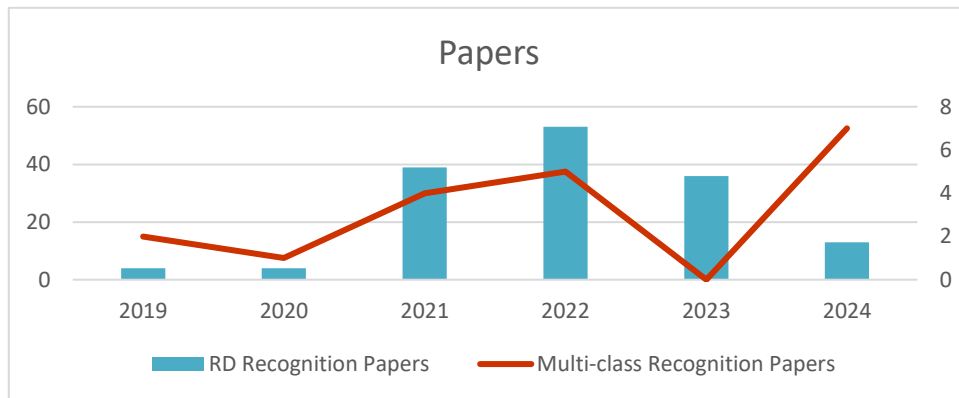


FIGURE 2. The number of studies on multi-class RD diagnostic and RD diagnostic in recent years.

Figure 2 delineates the temporal evolution of research paradigms in acoustic-based respiratory diagnostics, comparing single-disease recognition approaches with multi-class classification methodologies. The analysis reveals a significant shift towards multi-class frameworks, particularly accelerating post-2020, with notable peaks in 2022 and 2024. This trend reflects the field's progression toward more sophisticated diagnostic approaches, emphasizing comprehensive disease classification

capabilities through advanced artificial intelligence frameworks.

H. PAPER ORGANIZATION

This systematic review is structured into eight sections: introduction, datasets and data preprocessing, computational architectures, cough sounds features, classification and diagnostic methodologies, latest AI

technologies for respiratory disease recognition, discussion, and conclusion. As shown in Figure 3.

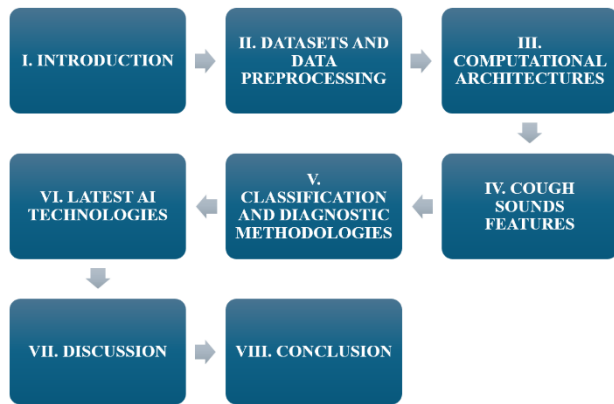


FIGURE 3. Schematic diagram of the overall structure of the article.

II. DATASETS AND DATA PREPROCESSING

This section examines datasets and evaluation protocols for multi-class respiratory disease classification.

TABLE I

SUMMARY OF OPEN-SOURCE DATASETS.

Dataset	Size	Description	Primary Applications	References
General-purpose Sounds Datasets				
Google AudioSet	2M+ clips	Labeled 10-second YouTube audio segments across diverse categories	Audio classification Multi-modal analysis	[46], [31]
LibriSpeech	1000h	English speech corpus from audiobooks	Speech processing Audio analysis	[31]
ESC-50	2000 recordings	Environmental sound recordings in 50 classes	Sound classification	[33]
TASK	17,393 samples	6,000 cough and 11,393 non-cough recordings	Cough detection	[31]
Specific Pathological Cough Datasets				
Coswara	941 participants	Respiratory sound recordings with demographic metadata from COVID-19 patients and healthy controls	COVID-19 detection	[31], [30], [46],
	6507 clean audio 1117 noisy audio		Respiratory disease classification	[47], [5], [48], [32]
ComParE	517 subjects	Audio recordings from 119 COVID-19 positive and 398 healthy subjects	COVID-19 diagnosis Audio feature analysis	[31]

2) CLINICAL DATA COLLECTION

Clinical cough data collection is fundamental for the development and refinement of multi-class recognition and diagnostic models targeting respiratory diseases. The collection process adheres to rigorous clinical validation and standardized protocols, including confirmed diagnoses by certified physicians, controlled recording environments, calibrated equipment, high sampling rates (≥ 44.1 kHz), and

A. ACOUSTIC DATA ACQUISITION AND QUALITY CONTROL

1) OPEN-SOURCE REPOSITORIES

Open-source repositories provide valuable resources for respiratory disease research, with key considerations in metadata standards and data interoperability [42]. These repositories follow Findable, Accessible, Interoperable, Reusable (FAIR) principles to ensure research reproducibility [43]. Table I characterizes these repositories and their applications.

Such large-scale datasets are primarily used for model pre-training. The integration of multiple general and environmental sound datasets has proven effective in enhancing the robustness and generalization of cough detection models [44]. In parallel, specific pathological cough datasets have been developed as foundational resources for multi-class identification and diagnostic tasks [45].

the documentation of detailed clinical metadata.

The datasets encompass demographically diverse populations with various respiratory conditions, and undergo expert annotation for diagnostic validation [49]. These datasets support both binary and multi-class classification tasks. While binary classification focuses on specific disease pairs [50],[51], multi-class scenarios require enhanced model generalization capabilities [52].

The integration of diverse datasets enhances model robustness [51]. As summarized in Table II.

B. DATA PROCESSING

The systematic preprocessing of acoustic signals, encompasses adaptive normalization protocols,

sophisticated noise reduction algorithms, advanced segmentation techniques, and Data Stratification and Validation, which constitutes a critical methodological framework for optimizing signal quality and ensuring analytical compatibility.

TABLE II
SUMMARY OF CLINICAL DATA DATASETS.

Study	Disease Categories	Dataset Composition	Sample Size	Ref.
Binary Classification Studies				
Pediatric Assessment	Asthma vs. Pneumonia	Cough recordings Pertussis cough (n=100)	850	[51]
Mixed Source Analysis	Croup vs. Pertussis	Croup cough (n=100) Other types (n=100) Non-cough (n=100) Normal (n=220)	400	[32]
Cough Classification	Obstructive vs. Restrictive	Obstructive (n=304) Restrictive (n=292) Asthma cough (n=73)	796	[52]
Cough Classification	Asthma vs. COPD	COPD cough (n=73) Normal (n=73)	219	[53]
Large-scale and Multi-class Classification Studies				
Respiratory Diseases	COPD, Asthma, COVID-19	Cough samples across conditions COVID-19 (n=241)	348	[46]
Multi-condition	COVID-19, Heart Failure, Asthma, Healthy	Heart failure (n=244) Acute asthma (n=110) Healthy (n=247)	2,944	[5]
COVID-19 Study	COVID-19, Pneumonia, Asthma	COVID-19 positive Non-COVID (n=341) COVID-19 (n=471)	1,183	[47]
COVID-19 Analysis	COVID-19, Healthy, Asthma, Pneumonia	Healthy (n=1,176) Asthma (n=92) Pneumonia (n=38)	1,777	[48]
Comprehensive Chest Diseases	Multiple conditions	Imaging Data: CXR: 78,785 images CT: 72,209 scans Audio Data: 2,724 cough sounds 449 minutes total duration	150,994 images 2,724 sounds	[30]
Pediatric Respiratory Study	Multiple conditions	Training set (n=852) Testing set (n=585) Ages: 29 days to 12 years	1,437	[54]

Including: normal cold, tonsillitis, adenoids, laryngotracheo bronchitis, croup, asthma, pneumonia, COPD Disease categories include: COVID-19, Lung Cancer, Atelectasis, Colitis, TB, Pneumothorax, Edema, Pneumonia Training set composition: croup (n=50), bronchiolitis (n=102), RAD (n=193), pneumonia (n=72), URTI (n=121), LRTD (n=522), no disease (n=123)

1) NORMALIZATION PROTOCOLS

Normalization adjusts signal levels to a standardized range, addressing volume inconsistencies across recordings[55]. Key techniques include Amplitude-based normalization, which divides each signal point by an amplitude function

derived from lowpass filtering. Loudness-based normalization, adjusts the perceived loudness but may alter temporal morphological features [56]. Peak normalization, scales the signal based on its maximum amplitude [57]. Root Mean Square(RMS) normalization, standardizing the root mean square value to a predefined level [58]. While

effective for regulating dynamic ranges, normalization techniques may introduce non-linearities that interact with other processing tools, potentially influencing spectral analysis outcomes.

2) NOISE REDUCTION ALGORITHMS

Noise reduction frameworks implement adaptive filtering algorithms to eliminate acoustic interference patterns while preserving essential spectral characteristics and temporal integrity of the diagnostic signals. Common techniques include spectral subtraction, which operates in the frequency domain to attenuate stationary noise components [59]. It is particularly effective for enhancing speech signals corrupted by additive noise, especially in steady-state or slowly varying environments. Moreover, it can be extended to non-stationary contexts through multiplicative implementations [60]. The Least Mean Square (LMS) adaptive filtering algorithm dynamically adjusts its parameters to accommodate fluctuating noise conditions [61], making it well-suited for non-stationary noise environments, such as impulsive disturbances or time-varying interference. An advanced variant, Delayed LMS (DLMS), has been shown to outperform both LMS and Normalized LMS (NLMS) in suppressing Gaussian and impulsive noise, particularly in speech and cardiac signal processing applications [62].

3) SEGMENTATION PROCESSING:

Cough sound segmentation techniques include energy-based detection, deep learning, Hidden Markov Models (HMM), spectral entropy, and zero-crossing rate (ZCR). Energy-based methods, such as standard deviation and short-term energy, are computationally efficient and effective for silence removal but sensitive to noise [63]. Deep learning approaches achieve superior accuracy, but require large labeled datasets [64]. HMMs excel in temporal modeling, yet involve complex parameter estimation and convergence challenges. Spectral entropy adapts well to non-stationary signals but is parameter-sensitive, while ZCR is simple yet noise-prone [65]. In practice, hybrid methods combining multiple features and algorithms are often used to improve segmentation accuracy and robustness.

4) DATA STRATIFICATION AND VALIDATION

Stratified sampling and balanced data selection help address class imbalance issues and mitigate dataset-related biases [66], while preserving demographic and clinical diversity. Datasets are typically stratified based on class distribution. In most cases, the training set comprises 60–70% of the data, with validation and testing sets each accounting for 10–20%. For smaller datasets, K-fold cross-validation is employed to enhance the robustness of model evaluation. Recent studies advocate feature-aware splitting strategies, particularly for

imbalanced datasets, to ensure that critical feature distributions are preserved across data partitions [67]. Fold-stratified cross-validation also helps prevent data leakage caused by duplicated records, while maintaining privacy in federated learning environments [68].

C. MULTI-DATASET FUSION

1) MULTI-SOURCE HETEROGENEOUS DATA FUSION

Multi-source heterogeneous data fusion addressing challenges posed by diverse data sources with varying fidelity, formats, and parameters [69]. Including attention mechanisms for adaptive feature weighting and spectral enhancement [70], as well as multiscale feature extraction techniques, such as squeeze-and-excitation modules [71]. Other effective methods include encoder-decoder methods, co-training, latent variable modeling, graph neural networks, and generative neural networks for multimodal data fusion [72]. Nevertheless, these methods have limitations: attention mechanisms and cross-modal interactions suffer from high computational complexity and low interpretability; multiscale extraction and adaptive reconstruction are sensitive to noise and training instability; graph-based and generative models demand strict data structure and intensive computation; co-training and latent variable modeling are vulnerable to initial assumptions.

2) FUSION STRATEGIES IN DOMAIN ADAPTATION

Fusion strategies in Domain Adaptation (DA) are crucial for facilitating effective cross-domain knowledge transfer and enhancing the generalization capabilities of models across diverse target domains. Multi-Source Domain Adaptation (MDA) improves target adaptation by leveraging multiple source domains, while feature-level fusion demonstrates superior classification performance [73]. Source-Free Domain Adaptation (SFDA) addresses privacy concerns by using only pre-trained source models. Approaches such as Adversarial SOURCE GEneration (ASOGE) synthesize source-like data [74], while others employ feature alignment with uncertainty estimation and self-training [75]. Recent developments also include Conditional Adversarial Domain Adaptation (CDAN), Multi-Discriminator methods (MDD, MADA), Generative Domain Adaptation (CyCADA), and Transformer-based approaches (DaViT, TransDA). Although effective, these methods face challenges like discrimination degradation, instability, and high computational cost.

D. DATASET ANALYSIS AND CHALLENGES

1) COMPARATIVE ANALYSIS OF DATASETS

Open-source datasets, such as Google AudioSet and Coswara, provide comprehensive acoustic data suitable for

model pretraining and broad classification tasks [76], [29]. These repositories feature diverse acoustic environments and standardized sampling rates, facilitating robust model generalization. Conversely, clinical datasets offer precisely labeled pathological conditions under controlled recording environments [77], [78], particularly valuable for specialized applications including pediatric diagnosis and rare disease research [78]. In addition, the datasets demonstrate significant diversity in both scope and scale. Binary classification tasks range from pediatric condition differentiation [50] to specific disease pair discrimination [46], while multi-class frameworks address more complex diagnostic scenarios. Dataset sizes vary considerably, from focused collections of 173 subjects [50] to extensive repositories of 1,183 samples [47], reflecting the intrinsic limitations and heterogeneity inherent in clinical data collection processes.

2) TECHNICAL CHALLENGES

Despite the availability of several open-source and clinical datasets, significant challenges persist. Limited data coverage is evident in insufficient sample sizes, incomplete disease type representation, and lack of demographic diversity [79]. For instance, the significant gap remains in studies on rare respiratory conditions (e.g., pertussis, tuberculosis) and patient groups, including pediatric, geriatric, and comorbid populations, presenting significant challenges for robust model development [80], [81]. Moreover, demographic and geographical biases within existing datasets may compromise model fairness and limit clinical applicability across diverse patient populations, particularly when models are deployed in new settings [82]. Medical datasets frequently exhibit significant class imbalance, with common diseases and healthy samples often dominating the datasets, making it difficult to detect rare conditions. This imbalance reduces the classifier's ability to accurately identify minority classes, leading to biased predictions and misclassifications [83].

Data quality issues include inconsistent recording environments, varying equipment quality, and lack of standardized signal quality [84]. Annotation problems involve non-uniform standards, insufficient expert verification, and incomplete clinical information [85]. Additionally, the lack of standardized data coordination protocols, especially concerning crowdsourced and smartphone-recorded cough sounds, poses substantial challenges to data reliability and reproducibility [86]. Privacy and ethical concerns include inadequate protection of patient confidentiality, unclear data ownership, and potential exploitation of patient information for commercial interests [87]. Balancing innovation with ethical considerations is crucial to maintain patient trust and protect their rights [88]. These limitations significantly impact

dataset quality and usability, necessitating careful consideration in data utilization and model development.

E. EVALUATION FRAMEWORKS

1) BINARY CLASSIFICATION PERFORMANCE

METRICS

Traditional evaluation metrics encompass accuracy, precision, recall (sensitivity), specificity, F1-score, and AUC [89]. Accuracy measures overall correctness but can be misleading when sensitivity is low. Precision evaluates the accuracy of positive predictions, while recall assesses the classifier's ability to identify true positive cases. The F1-score balances precision and recall, and AUC measures the model's ability to distinguish between classes [90]. On imbalanced datasets, the Matthews Correlation Coefficient (MCC) is proposed as a superior alternative [91]. MCC outperforms Cohen's Kappa and Brier score in binary classification assessment. Nonetheless, [92] argue that MCC and F1 have limitations and suggest using "real accuracy" and Cohen's Kappa instead. Proper metric selection should be based on the dataset characteristics and model evaluation goals [93].

2) MULTI-CLASS EVALUATION STRATEGIES

Multi-class classification scenarios demand specialized evaluation frameworks to address the inherent complexities of multiple disease categories. Macro metrics, such as macro F1-score, calculate performance averages across all classes without considering class sizes, while weighted metrics give more weight to larger classes [94]. When dealing with class imbalance, alternative metrics such as AUC, balanced accuracy, and G-Mean can offer more reliable performance evaluations [95]. Furthermore, in the evaluation of correlation and agreement, Cohen's Kappa and the Matthews Correlation Coefficient (MCC) are widely used metrics for evaluating classification performance, particularly in multi-class settings. These metrics help address class imbalance and provide a more comprehensive evaluation.

3) CLINICAL VALIDATION AND STATISTICAL TESTS

In medical diagnostics, recall is often prioritized to minimize missed detections. Complete confusion matrices and confidence scores are also critical for accurate decision-making [90]. Additionally, generalized F1-scores with customizable weights and probabilistic extensions of traditional metrics offer enhanced flexibility for complex tasks [96]. Choosing the right metric depends on factors such as the application's goals, dataset balance, and the trade-offs between precision and recall. Domain-specific needs and the interpretability of results are crucial when

selecting evaluation criteria [97]. As summarized in Table III.

Statistical tests for evaluating model performance metrics offer more reliable and interpretable conclusions [98]. For

comparing multiple classifiers across datasets, non-parametric tests like the Wilcoxon signed-rank test and Friedman test are recommended [98].

TABLE III
SUMMARY OF EVALUATION METRICS.

Metric	Mathematical Definition	Suitable Scenarios	Advantages	Limitations
Multi-class Accuracy	$Multi-class Accuracy = \frac{\sum_{i=1}^N \delta(\hat{y}_i, y_i)}{N}$	Balanced multi-class Overall assessment	Simple interpretation Quick overview	Hides class-specific issues Biased to majority
Macro Precision	$Macro Precision = \frac{1}{K} \sum_{k=1}^K \frac{TP_k}{TP_k + FP_k}$	Equal class importance Rare conditions	Fair to all classes Minority sensitive	May undervalue majority Outlier sensitive
Macro F1 Score	$Macro F1 = \frac{1}{K} \sum_{k=1}^K \frac{2 \cdot Precision_k \cdot Recall_k}{Precision_k + Recall_k}$	Complex classification Multiple conditions	Balanced evaluation Comprehensive	Complex interpretation May overemphasize rare
UAR	$UAR = \frac{1}{K} \sum_{k=1}^K Recall_k$	Imbalanced data Fair comparison	Class balanced Imbalance robust	Small class sensitive May be optimistic

TP: True Positive TN: True Negative FP: False Positive FN: False Negative

n: number of classes K: The total number of classes in the classification problem. k: The index of the current class (e.g., the k-th class).

N_k : The number of samples in the k-th class. N: The total number of samples across all classes

III. COMPUTATIONAL ARCHITECTURES

This section provides an overview of the computational frameworks employed for multi-class respiratory disease classification. Table IV outlines representative architectural configurations adopted in existing research.

A. MACHINE LEARNING

1) SUPERVISED LEARNING (SVM, LR, LDA)

Traditional supervised learning models, like Support Vector Machines (SVM), Logistic Regression (LR) and linear discriminant analysis (LDA), are widely used for sound classification tasks by leveraging features such as Mel-frequency cepstral coefficients (MFCCs), short-time energy, and zero-crossing rate [99]. While these models excel at handling labeled data and performing precise classification, especially for smaller datasets and Embed real-time systems [50], [51], [52].

Multi-class Support Vector Machine (SVM) architectures have been developed to address complex classification problems efficiently [100]. For embedded systems, a streaming architecture for multi-class SVM classification

has been developed, enabling continuous data processing without throttling input streams [101].

Multi-class logistic regression models with L1 penalties offer effective solutions for classification and variable selection in high-dimensional settings. The model penalizes differences between class-specific parameter vectors, improving interpretability and performance [102].

Linear Discriminant Analysis (LDA) effectively distinguishes cough sounds in multi-class classification tasks. Its probabilistic variant (PLDA) demonstrates superior performance compared to traditional classifiers in respiratory disease detection[103].

2) INSTANCE-BASED LEARNING (KNN)

Instance-based learning methods, such as K-Nearest Neighbors (KNN) and Clustered K-Nearest Neighbors (CKNN), offer an intuitive approach to classification by relying on similarity measures and distance calculations between data points [52], [47], [104]. Multi-class classification approaches have been successful in differentiating various lung diseases, including asthma, COPD, and pneumonia, with KNN classifiers achieving up to 99.81% accuracy using spectral features [105].

TABLE IV
SUMMARY OF MACHINE LEARNING MODELS AND DEEP LEARNING MODELS

Category	Model Architecture	Key Features	Applications	References
Machine Learning Models				
Supervised Learning	SVM	Kernel-based classification	Binary disease classification	[50], [51], [52], [106]
	Linear	Regularized optimization	Pediatric respiratory diagnosis	
	Discriminant Analysis (LDA)	Statistical modeling		[47]
		Distribution-based		

Instance-Based Learning	KNN	Distance-based learning	Disease probability estimation	[47], [52], [106]
	Classifier KNN	Non-parametric approach	Risk assessment	
Ensemble Learning	Random Forest (RF)	Multiple classifier fusion	Cough pattern recognition	[107], [106]
		Bagging and boosting	Disease similarity analysis	
Deep Learning Models				
CNNs	CNN	Deep feature learning	Multi-class disease detection	[31], [32], [52], [53], [108],
	ResNet (18/34/50)	Transfer learning	Real-time diagnosis	[30], [31], [32], [48],
	Inception V3	Attention mechanisms		[30], [32],
	MobileNet V2			[32],
	VGG19			[30], [46]
	Attention CNN			[30]
RNNs	LSTM	Temporal modeling	Time-series analysis	[31], [32], [50], [52], [53]
	CNN-LSTM	Hybrid architectures	Disease progression	[52], [31], [53]
Specialized Networks	Listen2Cough	Disease-specific design	Multi-disease classification	[33]
	DMRNet	Custom feature extraction	Clinical implementation	[109]
	DKPNet41	Real-time processing		[5]

3) ENSEMBLE LEARNING (RF)

Random Forest (RF) is an ensemble learning algorithm that enhances multi-class classification accuracy and robustness by constructing multiple decision trees. It has demonstrated high efficacy across a range of domains, including cough sound analysis for the detection of chronic obstructive pulmonary disease (COPD) and congestive heart failure (CHF) [106], as well as biomedical text classification for cancer diagnosis [110]. Recent advances have integrated interval temporal logic into RF and decision tree frameworks, yielding improved accuracy and interpretability in classifying cough and breath sounds [111]. Additionally, RF provides internal estimates of generalization error and variable importance through out-of-bag samples, further strengthening its applicability in real-world scenarios [112].

B. DEEP LEARNING FRAMEWORKS

1) CNNs

In deep learning, Convolutional Neural Networks (CNNs) and their variants (e.g., multi-layer CNNs, channel attention CNNs, the ResNet series, MobileNet v2, etc.), transforming audio signals into spectrograms, which allows them to extract time-frequency features, for efficient classification. Such as, Braided CNNs, inspired by auditory cortex neurons, demonstrate high recognition accuracy. ResNet addresses the vanishing gradient problem through residual connections [113], with AURO ResNet achieving state-of-the-art results in COVID-19 detection from respiratory sounds [114]. MobileNet series enhances efficiency through depthwise separable convolutions and inverted residual structures [115]. VGG19 shows strong performance in

audio processing, particularly in speech-related tasks [116], with transfer learning enabling effective adaptation to specific sound classification problems. Channel-Attention Dense U-Net [117] and spatial-channel attention mechanisms [118] demonstrate significant improvements in audio processing tasks.

2) RNNs

Long Short-Term Memory (LSTM) networks, an advanced variant of recurrent neural networks (RNNs), address the vanishing gradient problem in traditional RNNs through memory units and gate mechanisms, enabling the modeling of long-term dependencies over thousands of time steps. Hybrid CNN-LSTM models combine the spatial feature extraction capabilities of Convolutional Neural Networks (CNNs) with the temporal dependency modeling of LSTMs, have demonstrated high accuracy in detecting respiratory pathologies [31], [52]. Additionally, BiLSTM-CNN models have been effective in time-series analysis and disease severity assessment.

3) SPECIALIZED NETWORKS

Specialized neural architectures have been developed to address the unique challenges of multi-class respiratory disease classification through cough sound analysis. These networks incorporate advanced features:

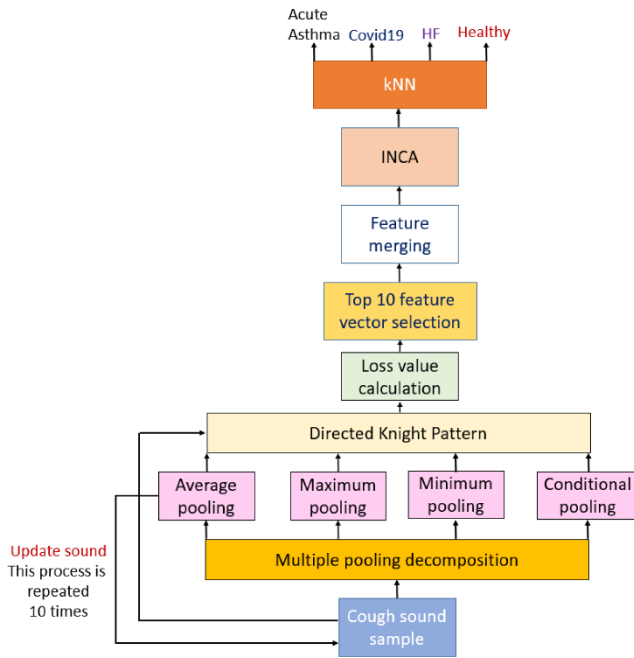


FIGURE 4 Schematic of the directed knight pattern-based model DKPNet41.

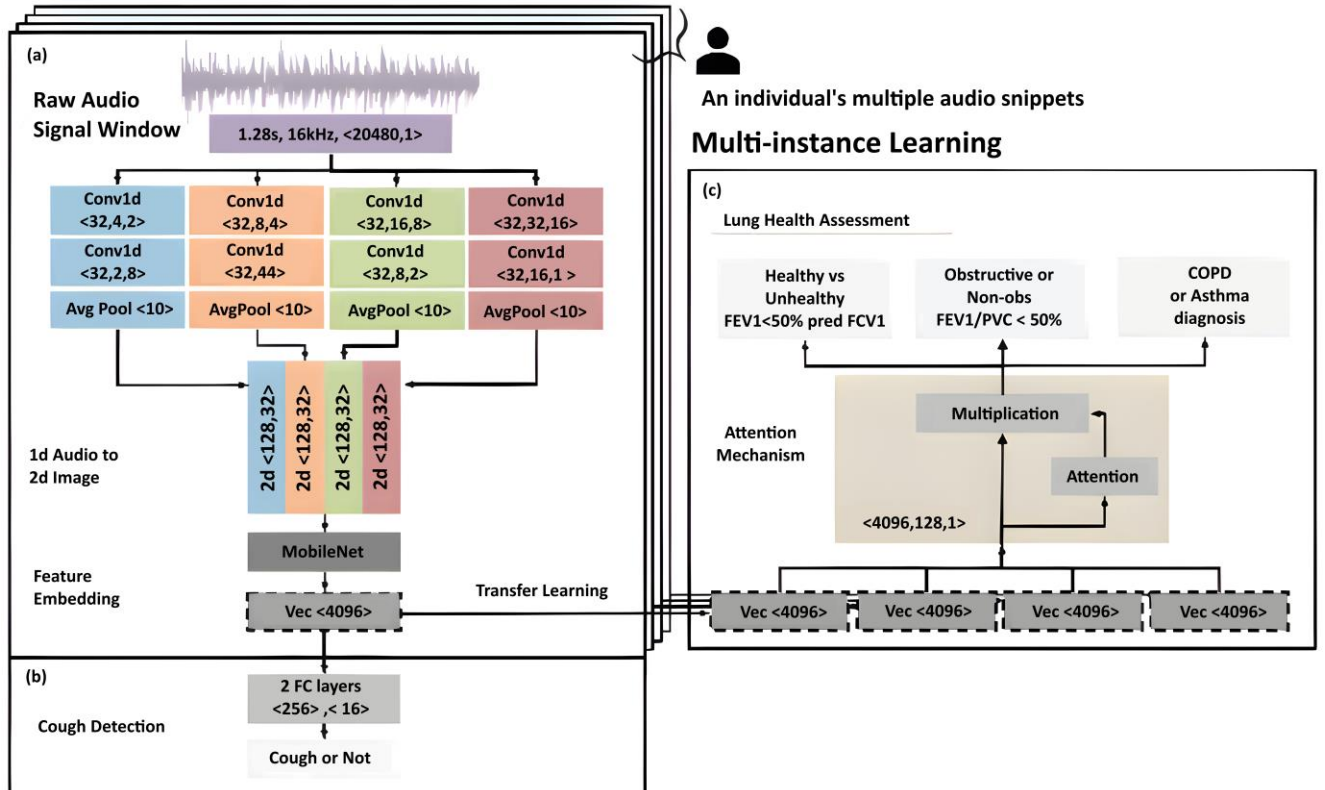


FIGURE 5. The overview of the framework of Listen2Cough.

DKPNet41: The model consists of four main components: novel feature generation based on a specifically directed knight pattern (DKP), signal decomposition using four pooling methods, feature selection using iterative neighborhood analysis (INCA), and classification using the k-nearest neighbor (kNN) classifier with ten-fold cross-validation [5]. As shown in Figure 4.

DKP is a local textural feature generator that generates 1,536 features per run. INCA combines One-shot Similarity learning and Neighbourhood Components Analysis to improve distance metrics for feature [119]. Multilevel Multiple Pooling performs pooling operations across different network layers, effectively integrating fine-grained and coarse-grained semantic information[120].

Listen2Cough: End-to-end deep learning with pre-trained MobileNet for feature extraction. Multi-instance learning framework for comprehensive cough analysis. Attention mechanism for identifying health-critical cough patterns. Demonstrates strong generalizability across diverse datasets [33]. This model framework utilizes two 1D-CNN layers to capture both temporal and spectral information from the

audio. To extract features at different temporal resolutions, an Inception-like module is employed. The resulting 2D feature map is then fed into a pre-trained MobileNet model, and the fine-tuned cough detection model is integrated into a Multi-Instance Learning (MIL) framework with an attention mechanism. As shown in Figure 5.

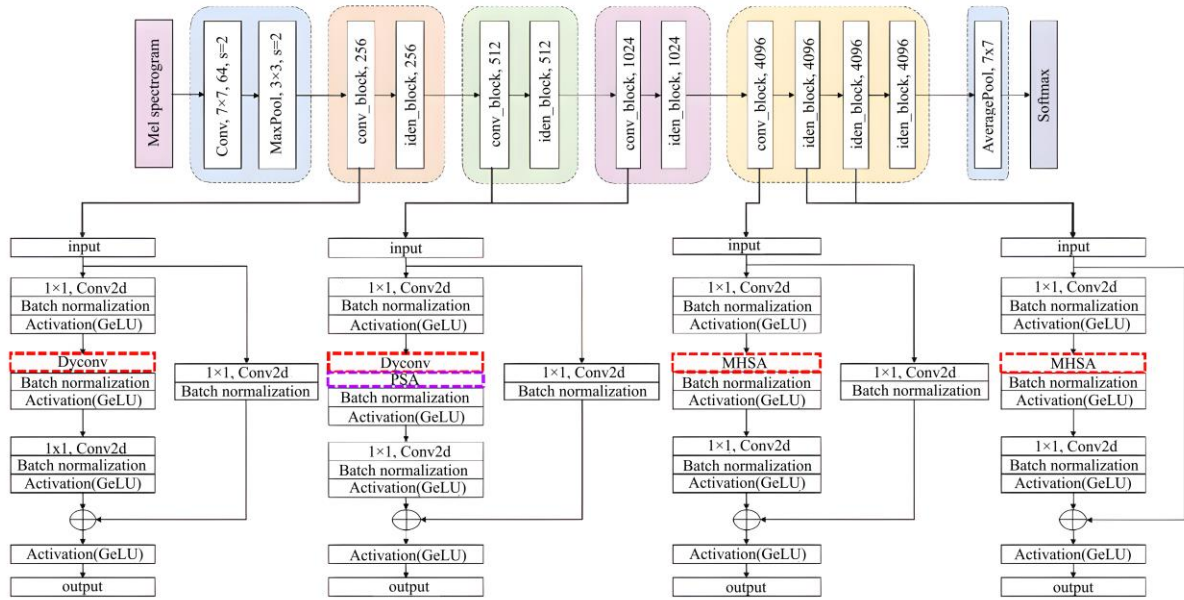


FIGURE 6. Overall framework diagram of DMRNet model.

DMRNet: Streamlined ResNet design with optimized block structure[109]. The DMRNet framework primarily consists of four convolutional blocks and six identification blocks. In addition, the model architecture integrates three key modules: The Dynamic Convolution Module introduces an efficient method for calculating dynamic convolution kernels through an attention mechanism [121], enabling the model to dynamically adjust its convolution operations based on the input. Polarized Self-Attention (PSA) is an attention mechanism module designed to improve pixel-wise mapping tasks in computer vision [122]. Multi-head self-attention (MHSA) enables models to capture long-range dependencies and global contextual information across spatial positions. As shown in Figure 6. These specialized architectures significantly advance multi-class respiratory disease diagnosis by: Handling class imbalance in respiratory datasets, extracting disease-specific acoustic features, Supporting real-time clinical applications. Figures 4-6 illustrate the architectural frameworks.

IV. COUGH SOUNDS FEATURES

This section introduces feature extraction, include time-domain features that capture temporal dynamics, frequency-domain features representing spectral composition, and advanced representations such as time-frequency analyses, higher-order statistical attributes, and Clinical feature.

A. BASIC FEATURES

1) TIME-DOMAIN FEATURES

Time-domain features are commonly leveraged to extract characteristics from sounds and other temporal signals, providing insights into instantaneous changes and overall signal distribution[123]. Key metrics include: Zero Crossing Rate (ZCR) measures the frequency of signal polarity changes, reflecting noise levels and short-term activity occurrence [124]. Root Mean Square Energy (RMSE), which captures amplitude envelope variations and

energy dynamics. Short-time energy detects amplitude variations and is essential for voice activity detection [124]. Time-domain features providing complementary measures that aid in the detection of voiced and unvoiced segments in speech signals. Furthermore, making them computationally efficient and suitable for real-time systems is critical for cough recognition.

2) FREQUENCY-DOMAIN FEATURES

Frequency domain features are obtained by applying Fourier transformation to the signal, extracting the frequency domain information of the signal [123]. Key metrics include: 1) Spectral Centroid (SC) indicates the brightness of the signal by calculating the weighted center of the frequency spectrum. 2) Spectral Flatness Measure (SFM) quantifies how "flat" or noisy the spectrum is, helping to differentiate between harmonic content and noise [125]. 3) Spectral Roll-off measures high-frequency content distribution [126]. 4) Cepstral peak prominence smoothed (CPPS) is effective in distinguishing healthy from pathological voices, correlating strongly with overall dysphonia grade and breathiness.

B. ADVANCED FEATURES

1) TIME-FREQUENCY REPRESENTATIONS

Time-frequency analysis combines both domains to provide a comprehensive understanding of audio signals, tracking how frequency content changes over time. Various transforms, such as Discrete Fourier Transform (DFT), Discrete Sine Transform (DST), and Discrete Cosine Transform (DCT), can convert time-domain signals into frequency-domain representations [127]. Spectrograms and mel-spectrograms are commonly used, but other representations like scalograms and gammatonegrams have also shown promise. Texture analysis of time-frequency images, particularly using gray-level co-occurrence matrices, has demonstrated robustness to noise compared to traditional cepstral features [128]. Deep learning approaches, such as convolutional neural networks, have been successfully applied to these time-frequency representations for classification tasks. with some studies finding improvements using cochleagram images over traditional spectrograms [128]. Table V is shown below:

TABLE V
ADVANCED FEATURE EXTRACTION FRAMEWORK FOR RESPIRATORY SOUND ANALYSIS

Feature Category	Mathematical Definition	Physical Interpretation	Clinical Applications
Time-frequency Features			
Mel Spectrogram	$Mel\ spectrogram(x) = Filterbank\ transform(x)$	Perceptual frequency mapping	Disease pattern recognition
		Auditory system modeling	Acoustic signature analysis
		Energy band distribution	Temporal evolution tracking
Spectrogram	$Spectrogram(x) = \mathcal{F}(x(t)) ^2$	Time-frequency representation	Respiratory event detection
		Energy distribution	Disease progression monitoring
		Temporal dynamics	Pattern change analysis
Statistical Features			
Kurtosis	$Kurtosis(x) = \frac{1}{N} \sum_{n=1}^N \left(\frac{x_n - \mu}{\sigma} \right)^4 - 3$	Distribution peak sharpness	Abnormal event detection
		Outlier presence	Pattern characterization
		Extreme value frequency	Disease classification
IQR	$IQR = Q_3 - Q_1$	Central spread measure	Pattern stability analysis
		Robust variability indicator	Outlier detection
		Distribution width	Data quality assessment
Skewness	$Skewness(x) = \frac{1}{N} \sum_{n=1}^N \left(\frac{x_n - \mu}{\sigma} \right)^3$	Distribution asymmetry	Pattern abnormality detection
		Shape characterization	Disease characterization
		Pattern deviation	Signal quality assessment

2) STATISTICAL PARAMETERS

Statistical analysis frameworks provide a comprehensive characterization of acoustic signal distributions and morphological properties through sophisticated mathematical descriptors. These metrics encompass: central tendency measures quantifying distribution centers, dispersion parameters capturing signal variability, and distribution shape characteristics evaluating asymmetry and

peakedness through skewness and kurtosis analysis [129]. The integration of advanced statistical measures, including quantile-based analytics and temporal statistical parameters [130]. By leveraging higher-order statistical features to capture the nonlinear characteristics of signals, including sample entropy, maximal Lyapunov exponent, and Hurst exponent, has shown promise in capturing complex airflow dynamics and internal morphological changes in wet coughs [131].

3) PATHOPHYSIOLOGICALLY RELEVANT ACOUSTIC FEATURES

The generation of pathological cough sounds requires high intrathoracic pressures and airstream velocities, achieved through deep inhalation, glottic closure, and dynamic airway compression [132]. Therefore, during feature extraction, phonation features, vocal-tract features and airway flow mechanics features [133] are typically analyzed separately.

Phonation features: Fundamental frequency (F0) corresponds to the rate of vocal [134]. Jitter quantifies frequency perturbations by measuring short-term fluctuations in F0 between consecutive glottal cycles [135]. Shimmer reflects glottal closure consistency [136]. The glottal-to-noise excitation ratio (GNE) estimates the proportion of periodic vocal-fold excitation versus turbulent noise excitation, demonstrated the best ability to distinguish between different pathological phonation mechanisms and normal voices [137]. The open quotient (OQ) representing the ratio of glottal opening time to total cycle duration [138]. Closed Quotient (CQ), derived from flow glottograms, represents the ratio of closed phase to period [139].

Vocal-tract features characterize the shape and resonance of the vocal tract. These include Linear Prediction Coefficients (LPC), Mel-Frequency Cepstral Coefficients (MFCC) [140], Perceptual Linear Prediction (PLP), Line Spectral Frequencies (LSF), spectral envelope, formants and their bandwidths, and band-power ratios of the power spectral density.

Airway flow mechanics features reflect pathological conditions such as airway obstruction, inflammation, collapse, and mucus accumulation. This including peak airflow, peak pressure, and expiratory rise time [141]. Among these, peak expiratory flow (PEF) has been identified as a valuable quantitative indicator of cough ability [142]. Cepstral peak prominence (CPP) has emerged as a reliable acoustic measure for assessing vocal quality and dysphonia [143].

C. FEATURE COMBINATIONS FOR SPECIFIC APPLICATION SCENES

Cough sound analysis leverages audio features from domains including automatic speech recognition (ASR) [144], acoustic scene analysis [145], sound event detection [146], Speaker Recognition [147], and deep learning-based representations [148]. Table VI is shown below:

To optimize feature combinations for multi-class cough recognition, integrate MFCC, Log-Mel, nonlinear features (kurtosis, entropy), and deep learning embeddings (VGGish, wav2vec 2.0). Use time-frequency masking and GFCC for noise robustness. Include cough-specific temporal features (e.g., attack/decay time) to distinguish cough types. Fine-tune CNN/RNN and self-supervised models on cough datasets. Validate on diverse cough datasets (e.g., asthma, COPD). These steps enhance multi-class cough analysis accuracy.

TABLE VI
FEATURE COMBINATIONS FOR SPECIFIC APPLICATION SCENES

Application Scene	Feature Combination	Role
ASR	MFCC, Perceptual Linear Prediction (PLP), Relative Spectral Filtering (RASTA), LPC, LPCC, Pitch, Voicing Features	Captures cough spectral characteristics, models vocal tract, supports fundamental frequency estimation, voice activity detection, and cough event segmentation for disease diagnosis and remote monitoring.
Acoustic Scene Analysis	Log-Mel Spectrograms, Constant-Q Transform (CQT), Gammatone Frequency Cepstral Coefficients (GFCC)	Provides robust time-frequency representations, mimics human auditory filtering, and enhances natural sound classification in noisy environments.
Sound Event Detection	Log-Mel Spectrograms, MFCC, Time-Frequency Masking, Sparse Coding, High-Order Statistics (Kurtosis, Skewness), Entropy Measures	Isolates cough events from background noise, captures nonlinear transient patterns, and improves detection accuracy in noisy settings.
Speaker Recognition	Jitter, Shimmer, Harmonics-to-Noise Ratio (HNR), Formants, Formant Rates of Change	Assesses voice quality, diagnoses vocal disorders, captures vocal tract characteristics, and supports forensic speaker identification.
Deep Learning-Driven Feature Extraction	CNN/RNN Embeddings (VGGish, Open-L3, YAMNet), Self-Supervised Learning Features (wav2vec 2.0)	Extracts high-dimensional features from Log-Mel spectrograms or raw waveforms, enabling generalizable feature extraction for complex audio tasks like speaker recognition.

V. CLASSIFICATION AND DIAGNOSTIC METHODOLOGIES

Following the theoretical analyses of the preceding chapters, this section provides an in-depth exploration of the diagnostic and classification approaches for multiple respiratory diseases.

A. FEATURE-BASED CLASSIFICATION STRATEGIES

1) MFCCS FEATURE-BASED METHODS

Optimal feature selection significantly impacts classification performance. MFCC-based classification achieves 85-92% accuracy through careful feature

engineering: combining basic coefficients with Delta and Delta-Delta features enhances temporal pattern recognition [50], [51]. Spectrogram-based approaches further improve accuracy to 89-95% by focusing on disease-specific frequency bands (1400-1650 Hz) [46], [108]. as summarized in Table VII.

TABLE VII
SYSTEMATIC ANALYSIS OF MFCC-BASED METHODS FOR RESPIRATORY DISEASE CLASSIFICATION

Study	Disease Categories	Dataset Composition	Methodological Framework	Performance Metrics
[50]	Pediatric Bronchitis vs. Pneumonia	Total: 173 samples Acute asthmatic bronchitis (n=1) Acute bronchiolitis (n=13) Acute bronchitis (n=76) Lobar pneumonia (n=1) Bronchopneumonia (n=81)	Models: SVM, LSTM Features: MFCC, STE, SNR Augmentation: Time/pitch shifting Preprocessing: BIC, Log-MMSE	AUC (SVM): 0.92 AUC (LSTM): 0.93
		Total: 852 coughs Croup (n=50) Bronchiolitis (n=102) RAD (n=193) Pneumonia (n=72) URTI (n=121) LRTD (n=522) Healthy (n=123)	Platform: ResApp Health Models: SoftMax NN, TDNN Features: MFCC, Δ MFCC, Δ^2 MFCC Additional: ZCR, kurtosis	Asthma: 97% Pneumonia: 87% LRTD: 83% Croup: 85% Bronchiolitis: 84%
[51]	Pediatric Asthma vs. Pneumonia	Total: 850 recordings Pediatric patients	Model: SVM Features: MFCC Energy features	Accuracy: 90%
[52]	Obstructive Cough, Restrictive Cough, and Normal Cough	Total: 816 samples Normal (n=220) Obstructive (n=304) Restrictive (n=292)	Models: KNN, CNN, LSTM, SVM, CNN-LSTM Ensemble approach Feature: MFCC	Accuracy: 98.63% (Ensemble model)
[31]	Tuberculosis, COVID-19, Healthy	Total: 1,734 subjects TB (n=7) COVID-19 (n=229) Healthy (n=1,498)	Models: ResNet50 Features: MFCC, ZCR Enhancement: SMOTE Transfer Learning	F1 (TB vs COVID): 0.9259 F1 (3-class): 0.8631
[106]	COPD vs. CHF (Congestive Heart Failure)	Total: 36 subjects COPD (n=9) CHF (n=9) Controls (n=18)	Model: Random Forest Features: MFCC, ZCR, SPL Enhancement: SMOTE Validation: 10-Fold CV	Sensitivity: 82% Specificity: 75% Accuracy: 78.05% ROC-AUC: 0.80

2) SPECTROGRAM-BASED APPROACHES

Spectrogram-based approaches, particularly using Mel-spectrograms with advanced architectures, demonstrate superior performance in respiratory disease classification. VGG19 achieves ROC-AUC of 0.95 for COPD detection, while CNNs maintain 89% accuracy in three-way classification tasks [46], [108].

The effectiveness stems from optimized time-frequency analysis (1400-1650 Hz frequency bands) and multi-scale temporal pattern recognition. Transfer learning with pre-trained models enhances classification capabilities through

fine-tuning [47], [108]. In clinical applications, these methods achieve 89% accuracy and 0.90 AUC for pneumonia versus asthma discrimination [47], enabling robust integration into automated screening systems. Table VIII presents the methodological frameworks and performance metrics across implementations.

TABLE VIII
ANALYSIS OF SPECTROGRAM-BASED METHODS FOR MULTI-CLASS RESPIRATORY DISEASE CLASSIFICATION

Study	Disease Categories	Dataset Composition	Methodological Framework	Performance Metrics
[46]	COPD, Asthma, and COVID-19	Total: 348 samples Source: Coswara dataset Conditions: COPD, Asthma, COVID-19, Healthy	Architecture: VGG19 Approach: Transfer Learning Input: Mel-spectrograms Training: Fine-tuning	ROC-AUC by condition: COPD: 0.95 Healthy: 0.78 COVID-19: 0.70 Asthma: 0.67
[47]	Pneumonia, Asthma Coughs	Total: 1,524 samples COVID-19 positive (n=1,183) Non-COVID (n=341) Source: Coswara dataset	Models: LDA, KNN Analysis: PSD, Mann-Whitney U Feature: Frequency bands Optimization: Band selection	Accuracy: 89% AUC: 0.90
[108]	Bronchitis, Bronchiolitis, Pertussis	Total: 268 patients Bronchiolitis (n=35) Pertussis (n=131) Bronchitis (n=102)	Model: CNN Input: Mel-spectrogram Feature: Time-frequency Processing: Image-based	Accuracy: 89%

3) TIME-DOMAIN AND STATISTICAL METHODS

Time-domain features such as zero-crossing rate (ZCR) and energy-based metrics are widely used for analyzing temporal signal patterns and intensity, forming the foundation of many classification models[149]. Advanced statistical measures, including sample entropy, Lyapunov exponent, and Hurst exponent, quantify signal complexity, stability, and long-term dependencies, further enhancing classification accuracy [107]. Hybrid techniques integrating time-domain and spectral features demonstrate improved robustness and adaptability, as seen in frameworks like Listen2Cough [33]. Traditional models like CKNN and

neural pattern recognition excel in structured environments, while advanced frameworks, such as MobileNet and deep CNNs, process multi-feature inputs efficiently. Binary classification of COPD vs. asthma achieved an F1-score of 0.912 in clean environments, with minor drops under field conditions (F1:0.884) [33]. Multi-class applications, including eight-type pneumonia classification, highlight the scalability of these approaches across complex scenarios[149]. Despite their strengths, challenges such as sensitivity to noise, accuracy variations across conditions, and resource requirements remain critical areas for improvement [107]. as summarized in Table IX.

TABLE IX
ANALYSIS OF TIME-DOMAIN AND STATISTICAL FEATURE-BASED METHODS FOR RESPIRATORY DISEASE CLASSIFICATION

Study	Disease Categories	Dataset Composition	Methodological Framework	Performance Metrics
[33]	COPD and Asthma	Multiple sources: Google AudioSet ESC-50 Freesound ESTI/DEMAND/AIR Self-collected: 131 primary samples 70 validation samples	Architecture: Listen2Cough Base model: MobileNet Enhancement: 1D-CNN processing Adaptive resampling Environment simulation Noise augmentation Mixup strategy	Clean Data: F1: 0.912 Field Conditions: F1: 0.884
[149]	Eight Types of Pediatric Pneumonia Infections	Clinical dataset: 112 patient samples Pediatric population Verified diagnoses	Model: Deep CNN Feature extraction: Time-domain: ZCR, Energy Spectral: Centroid, Rolloff Statistical: Entropy	Test dataset accuracy: 0.4

		Acoustic: MFCCs, Chroma	
		Comprehensive analysis	
		Models:	
		CKNN	
		Neural Pattern Recognition	
[107]	Wheezing and Pertussis	Balanced dataset:	Accuracy: 94.7%
		Whooping (n=37)	
		Non-whooping (n=38)	
		Digital recordings	
		Advanced features:	
		Sample entropy	
		Lyapunov exponent	
		Hurst exponent	
		Non-linear analysis	

B. ADVANCED FEATURE CLASSIFICATION METHODS

1) MULTI-SCALE AND MULTI-LEVEL METHODS

Multi-scale and multi-level methods enable comprehensive characterization of disease-specific patterns by analyzing respiratory acoustic signals at different temporal and frequency resolutions. Table X summarizes these studies. These methods are highly applicable to clinical settings, supporting disease-specific tasks like respiratory infection detection and chronic condition monitoring, as well as broader public health challenges [32]. For binary tasks, such as distinguishing wheezing from pertussis, models achieved 90.5% accuracy and sensitivity, with a specificity of 93.92% [32]. In multi-class scenarios, such as four-class classification involving COVID-19, asthma, heart failure, and healthy subjects, models reached an impressive accuracy of 99.39% [5]. Advanced approaches also yielded an AUC of 0.74 across multiple respiratory conditions, underscoring their effectiveness in complex disease discrimination [48].

Shen et al. [48] proposed a framework incorporating maximum overlapping mel-spectrograms that maintains high temporal resolution through optimized window overlap strategies while preserving crucial frequency

information via mel-scale transformation. Figure 7 Schematic illustration of the multi-scale feature extraction workflow.

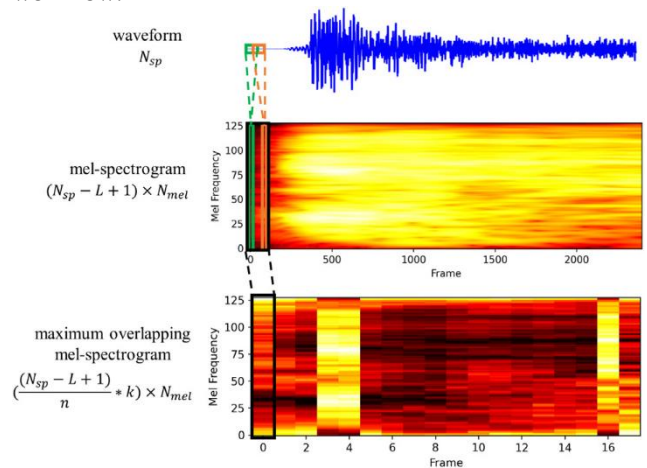


FIGURE 7. Schematic diagram for calculating maximum overlapping mel-spectrogram.

These methods enhance accuracy, interpretability, and efficiency in tasks such as image quality assessment and semantic categorization [150]. However, they face challenges including high computational demands and increased model complexity

TABLE X
SUMMARY OF MULTI-SCALE AND MULTI-LEVEL ADAPTIVE FEATURE EXTRACTION METHODS

Study	Disease Categories	Dataset Composition	Methodological Framework	Performance Metrics
[32]	Wheezing vs. Pertussis	Source: YouTube	Architecture:	
		Validated cough recordings	CNN with Channel Attention	Accuracy: 90.5%
		Clinical verification	Feature Extraction:	Sensitivity: 90.5%
			Multi-scale Mel spectrograms	Specificity: 93.92%
			Adaptive scale selection	
[48]	COVID-19, Healthy, Asthma, Pneumonia	Total: 1,777 subjects	Enhancement:	
		Distribution:	Channel attention mechanism	
		COVID-19 (n=471)	Model: ResNet18	
		Healthy (n=1,176)	Feature Engineering:	AUC: 0.74
		Asthma (n=92)	Maximum overlapping mel-spectrograms	
		Data Augmentation:		

[5]	COVID-19, Heart Failure, Acute Asthma, and Healthy	Pneumonia (n=38) Source: Coswara dataset	SpecAugment Noise addition Pitch/Time shifting Time stretching Self-energy enhancement Architecture: DKPNet Components: DKP feature generation KNN classification INCA optimization Enhancement: Multiple pooling decomposition Adaptive feature selection	Accuracy: 99.39%
		Total: 642 samples Conditions: COVID-19 Heart Failure Acute Asthma Healthy		

2) MULTI-MODAL FEATURE FUSION

Feature fusion strategies effectively capture cross-modal relationships while preserving modality-specific characteristics. The integration of clinical images (CXR, CT) and respiratory sounds enables comprehensive disease characterization, achieving superior performance in multi-class classification tasks [30]. Figure 8 illustrates this process.

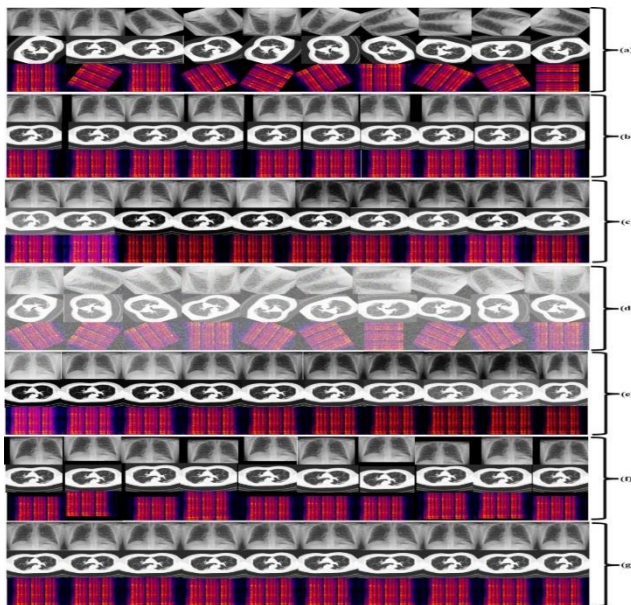


FIGURE 8. Generating CXR, CT scan, and CSI using multiple-way data generation (MWDG). Multi-modal data augmentation and generation were performed using MWDG techniques to enhance data diversity and mitigate overfitting during model training.

The MWDG framework implements seven key transformations for multi-modal data integration: (a) Rotation for orientation diversity, (b) HST and (c) VST for spatial transformations, (d) NI for noise injection, (e) GCOR for global correlation enhancement, (f) RTS for real-time scaling, and (g) Scaling for dimension adjustment [30]. Clinical implementation of feature fusion methods

emphasizes real-time diagnostic capabilities and integration into existing workflows.

3) DYNAMIC RESOLUTION NETWORKS

Dynamic Resolution Networks (DRNs) enhance respiratory disease classification through adaptive processing of varying image resolutions, addressing the limitations of fixed-resolution architectures. The network's dynamic nature allows it to process both fine-grained details in high-resolution regions and broader contextual information in lower-resolution areas, making it particularly effective for analyzing respiratory imaging data where both local and global features are crucial for accurate diagnosis. This adaptive processing strategy enhances the model's ability to identify subtle disease patterns while maintaining computational efficiency [109]. However, DRNs face challenges in implementation complexity and interpretability [151].

4) COMPLEX NETWORKS BASED ON GRAPH THEORY METHOD

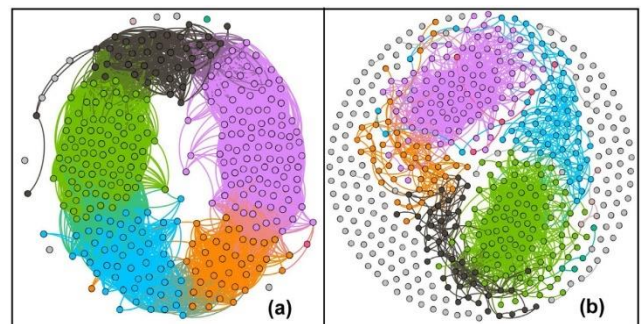


FIGURE 9. Complex networks based on graph theory of (a) croup cough and (b) pertussis. Centrality identifies the most critical nodes that play a significant role in the network's structure, and path length captures the temporal relationships between different features. The edge count quantifies the density of connections within the network, while graph density measures the overall connectivity of the network.

The complex networks based on graph theory to unwrap the hidden characteristics of cough signals, croup (BC), and pertussis (PS). The spectral and complex network analyses of 48 cough sounds are utilized for understanding the airflow through the infected respiratory tract. The spectral analyses reveal high-frequency components in both cough signals with an additional high-intense low-frequency spread in BC. The complex network features created by the correlation mapping approach, like number of edges (E), graph density (G), transitivity (Tr), degree centrality (D), average path length (L), and number of components (Nc) distinguishes BC and PS. The edge count quantifies the density of connections within the network, while graph density measures the overall connectivity of the network. Centrality identifies the most critical nodes that play a

significant role in the network's structure, and path length captures the temporal relationships between different features. These features capture structural properties and relationships within the graph, facilitating the learning of both local and global signal characteristics.

The extracted features undergo modular clustering to identify major clusters, which correspond to distinct signal feature regions. The network is visualized using Gephi 0.9.2 software (Figure 9), with the Fruchterman-Reingold layout algorithm optimizing the graph's layout. This visualization aids in understanding the distribution of different signal types and their interconnections.

The use of complex network methods in cough sound analysis reveals intricate patterns and correlations that traditional methods may overlook. As shown in Table XI.

TABLE XI
ADVANCED METHODOLOGICAL FRAMEWORKS

Study	Disease Categories	Dataset Composition	Methodological Framework	Performance Metrics
Multi-modal Feature Fusion				
[30]	COVID-19, LC, COL, Atelectasis (ATE), TB, Pneumothorax (PNEUTH), Edema (EDE), Pneumonia (PNEU)	Coswara Dataset: COVID-19+ (n=92) Healthy (n=1,079) Sarcos Dataset: COVID-19 (n=18) Healthy (n=26) Additional Data: 24 public chest datasets Multiple modalities (CXR, CT, CSI)	Deep Learning Models: VGG-19, ResNet (50/101) DenseNet (121/201) EfficientNetB0 Inception-V3 Enhancement Techniques: Scale Map Technique BANL, RBAP, MWDG SMOTE Tomek balancing	Overall Accuracy: 99.01%
Dynamic Resolution Networks				
[109]	Tuberculosis vs. Other Respiratory Diseases	Total: 1,323 samples Distribution: TB patients (n=141) Other diseases (n=52) Healthy subjects (n=152) Clinical validation	Architecture: DMRNet Components: Convolutional blocks Recognition blocks Self-attention mechanism Input: Mel spectrograms	Accuracy: 94.32% Sensitivity: 97.73% Specificity: 99.43%
Complex networks based on graph theory method				
[152]	Pertussis vs. Wheezing	Total: 48 samples Distribution: Pertussis (n=21) Wheezing (n=27) Clinical verification	Models: LDA classifier quadratic support vector machine (QSVM) Features: Spectral analysis Network characteristics Graph theory-based representation	LDA Accuracy: 94.11%

C. COMPARATIVE ANALYSIS OF METHODOLOGICAL APPROACHES

A systematic evaluation of respiratory disease classification methods highlights significant differences in performance, computational efficiency, and clinical applicability.

1) PERFORMANCE METRICS

In terms of classification accuracy, different methods exhibit a clear performance gradient. MFCC-based methods:

Achieve 85–92% accuracy in binary classification and 83–97% in multi-class tasks but are constrained by feature complexity [50], [51], [54]. Spectrogram-based approaches: Yield 89–95% accuracy by leveraging deep learning, offering superior pattern recognition and handling of complex acoustic features [108], [46]. By contrast, Feature fusion methods: Deliver the highest accuracy (99.01%), demonstrating robustness and enhanced performance in multi-class settings through multi-modal integration [30]. As shown in Table XII.

TABLE XII

COMPARATIVE ANALYSIS OF FEATURE EXTRACTION FOR MULTICLASS RESPIRATORY DISEASE DIAGNOSIS

Method	Binary Classification Accuracy	Multi-Class Classification Accuracy	Key Characteristics	References
MFCC-based Methods	85–92%	83–97%	Constrained by feature complexity, effective but limited in handling complex features	[50], [51], [54]
Spectrogram-based Approaches	89–95%	Not specified	Leverages deep learning, superior pattern recognition, handles complex acoustic features	[108], [46]
Feature Fusion Methods	Not specified	99.01%	Highest accuracy, robust, enhanced performance via multi-modal integration	[30]

2) IMPLEMENTATION CHALLENGES

In terms of resource requirements and deployment considerations, different methods exhibit significant disparities in their demands on hardware infrastructure. MFCC-based approaches can operate efficiently on standard computing systems, whereas spectrogram-based methods typically require GPU support, and multi-modal fusion techniques necessitate high-performance computing infrastructure. Notably, all approaches demand robust preprocessing capabilities to address environmental noise and ensure reliability in real-world scenarios. Concerning integration into clinical practice, incorporating AI systems into existing healthcare workflows requires a comprehensive consideration of multiple dimensions, including model optimization, system scalability, and healthcare staff training. The adoption of phased deployment strategies, combined with continuous performance validation and feedback mechanisms, is crucial for ensuring sustainable system implementation.

methods [154]. [155] uses contrastive pre-training and random masking to improve performance with limited labeled data. [156] uses wav2vec2.0 for self-supervised feature extraction on unlabelled audio. [157] employs bootstrap-based SSL and a novel Utterance-Frame Objective to pre-train on unlabeled audio, achieving approximately 15 times faster pre-training than prior SSL models.

VI. LATEST AI TECHNOLOGIES FOR RESPIRATORY DISEASE RECOGNITION

This section systematically analyzes seven transformative technological developments that are reshaping respiratory disease detection through innovative methodological frameworks. As shown in Figure 10.

A. SELF-SUPERVISED LEARNING

Self-supervised learning (SSL) is an unsupervised learning approach that leverages unlabeled data to learn representations for downstream tasks [153]. Four categories of pretext tasks: context-based methods, contrastive learning (CL), generative algorithms, and contrastive generative



FIGURE 10. Framework Diagram of Latest AI Technologies for Respiratory Disease Recognition

B. FEDERATED LEARNING

Federated learning (FL) enables collaborative learning across multiple healthcare systems without sharing sensitive patient data, enhancing privacy and security [158]. Studies have applied FL to various data types, including cough sounds [159], chest CT images [160], and chest X-rays [161]. Genetic clustered FL (Genetic CFL) has been proposed to groups edge devices based on the hypertuned parameters and modifies the parameters cluster wise genetically [162]. This approach was further applied to multi-modal COVID-19 diagnosis, achieving improvements of 16% and 11% in F1-

scores on X-ray and ultrasound datasets, respectively [163]. CoviFL, an edge-assisted FL framework for COVID-19 detection using cough audio, attaining 93.01% accuracy [164].

C. EXPLAINABLE AI

Explainable AI techniques like SHapley Additive exPlanations (SHAP), Local Interpretable Model-agnostic Explanations (LIME), Gradient-weighted Class Activation Mapping (Grad-CAM), and Layer-wise Relevance Propagation (LRP) can enhance transparency and interpretability of AI models in healthcare [165], [166]. In tasks that classify multiple pulmonary diseases, SHAP and LIME have been applied to generate saliency maps, helping clinicians understand which spectral components drive the classification [167]. CoughLIME enhances the LIME method for COVID-19 cough classification by incorporating Non-negative Matrix Factorization (NMF) to decompose input audio into human-interpretable components [168]. Other approaches combine Vision Transformers and Grad-CAM to generate heatmaps over cough spectrograms, enhancing transparency in COVID-19 diagnosis [19]. In addition, SmoothGrad and Grad-CAM have been used to visualize key spectral patterns in cough-based classification models [169]. For asthma detection, a study employed ResNet18 with attention mechanisms and applied Grad-CAM to lung sound recordings, enabling clear differentiation between pathological and healthy sounds [170].

D. TRANSFER LEARNING

Transfer learning is closely aligned to multi-task learning, allowing transfer across data distributions (domain adaptation), model architectures, and learning paradigms. For instance, [171] developed expert classifiers utilizing deep neural-based cough representation extractors (e.g., CNN or LSTM), which achieved high classification accuracy and robustness across multiple respiratory disease datasets. Similarly, [172] demonstrated that transfer learning and bottleneck feature extraction improved COVID-19 detection performance and generalization, with coughs showing the strongest COVID-19 signature. To reduce dependence on large patient datasets, an incremental transfer learning approach was introduced in [173], utilizing models pre-trained on healthy coughs to detect COVID-19-related coughs. Moreover, a cross-dataset transfer learning method was proposed in [174], incorporating cough detection, segmentation, and data augmentation, which significantly improving COVID-19 detection performance.

E. COMBINATION WITH QUANTUM COMPUTING

Quantum Support Vector Machines (QSVM) utilize quantum feature maps and kernels to manipulate quantum states for efficient classification tasks. QSVM's superiority over classical SVM in binary and multi-class classification

problems, with improved accuracy and reduced computational complexity [175], [176]. In aerodynamic classification, QSVM outperformed classical SVM by 11.1% in binary tasks and 17.9% in multi-class scenarios [176]. leverage quantum algorithms like HHL to achieve exponential speedup in computational complexity compared to classical SVMs [177], demonstrate enhanced performance in managing high-dimensional feature spaces [178], outperforming classical models in speech classification tasks with limited datasets [179].

F. MOBILE HEALTHCARE APPLICATIONS

Mobile healthcare platforms have revolutionized real-time respiratory monitoring and diagnosis, demonstrating remarkable adaptability in resource-limited environments. Real-time Detection Systems: Automated monitoring technologies have significantly enhanced real-time respiratory disease diagnosis. MobiCough implements GMM-UBM frameworks with Bluetooth integration for efficient monitoring [180], CASCO platform combines STFT-CNN algorithms with WeChat for real-time detection [181], and Hyfe Cough Tracker provides continuous cough monitoring across various environments [182]. COVID-19 Specialized Applications: Multiple COVID-19-specific diagnostic platforms have substantially improved diagnostic accuracy and real-time monitoring capabilities. AI4COVID-19 achieved 95.6% diagnostic accuracy through transfer learning [183], QUCoughScope provides optimized real-time detection frameworks enhancing response speed and diagnostic precision [184], while COVCOUGH integrates continuous monitoring functionality, effectively supporting epidemic control [185].

G. WEARABLE TECHNOLOGY INTEGRATION

Wearable devices, through sensor and edge computing technology integration, expanding the boundaries of telemedicine. A flexible multimodal sensor patch integrated with smartphone-based edge computing was developed for real-time monitoring of ECG, respiration, skin temperature, and cough signals [186]. Similarly, a wearable sensor system captures snapshot of physiological data, including cardiac, respiratory, temperature, and percent oxygen saturation [187]. Hearables such as HearCough and EarCough utilize active noise cancellation microphones to achieve continuous, low-power cough detection with high accuracy [188],[189]. Meanwhile, embedded systems using TinyML on Arduino devices in distinguishing coughs from background noise [190]. Smartwatches offer convenient respiratory monitoring [191], while advanced microphone arrays enhance acoustic fidelity in complex environments [192]. Moreover, smart masks and wearable sensors provide round-the-clock health monitoring [193].

VII. DISCUSSION

This section discusses key issues in data representativeness, model validation, regulation and ethics, evaluation metrics and interpretability, analyzes the gap between research and practice, and proposes future research directions.

A. DATA CHALLENGES AND REPRESENTATION BIAS

The Pathological cough dataset lacks diversity, focuses on specific regions or populations, and ignores pediatric, geriatric, and comorbid patients. The variability of datasets and the lack of global representation introduce selection bias [194], and the labeling relies on self-reported used as ground truth or confirmed cases causing subjectivity and response biases [195], and the model is prone to overfitting due to limited sample diversity. This bias can lead to inequitable outcomes in societal applications and undermine the effectiveness of data-driven algorithms [194]. Furthermore, data noise can decrease accuracy by up to 15% [196]. Inconsistencies in data collection protocols and formats have hindered progress [197]. Data harmonization standards are needed to improve the reproducibility and scalability of crowdsourcing and smartphone cough data. Future Directions: Blockchain-based distributed data collection, or combined with transfer learning to solve the lack of diversity. Build globally representative datasets that cover diverse populations and disease types. Develop data harmonization standards to optimize crowdsourced or smartphone recording data for reproducibility. Develop efficient data augmentation techniques to reduce computing requirements.

B. REGULATORY AND ETHICAL

Regulatory bodies like the FDA in the US and CE in Europe classify AI solutions (including cough diagnostic tools) as Software as Medical Device (SaMD) and require extensive evidence of safety, efficacy, and quality control [198]. The regulatory requirements vary based on the device's risk level and include administrative, software-related, clinical, and non-clinical evidence [198]. To address the unique characteristics of AI, the FDA has proposed an innovative framework adopting a total product lifecycle approach [199]. However, the lack of standardized guidelines for non-harmonized datasets remains a challenge in meeting regulatory requirements [200].

Moreover, integration of AI in healthcare presents significant privacy challenges, patient data sharing is essential for scientific validation and information reuse, but it raises concerns about privacy protection and data security [201]. To address these issues, researchers recommend implementing privacy-preserving technologies, such as encryption, anonymization, and federated learning [202]. Additionally, the development of explainable AI frameworks and dynamic consent models is crucial to ensure patients understand how their data is used and decisions are made

[202]. Balancing privacy preservation with data utility, requiring interdisciplinary collaboration and responsive governance to align healthcare AI with ethical principles and prioritize patient-centered outcomes [203].

Future Directions: Develop guidelines for standardizing AI models and provide clinically validated data to meet regulatory requirements. Privacy-preserving technologies such as federated learning are used to ensure data security and ethical compliance. Develop AI tools that interoperate with EHR systems and optimize interface design to fit clinical workflows.

C. INDICATORS FOR THE EVALUATION OF IMBALANCED MULTI-CATEGORICAL PROBLEMS

In the multi-classification task, the class imbalance can easily lead to the bias of the model towards the majority class, traditional accuracy metrics can be misleading in such scenarios, as they may conceal biases towards majority classes [204], and the macro F1 score and Cohen's kappa are more suitable for evaluation [205]. Compared with binary classification, multi-classification is more complex due to category overlap and fine-grained discrimination. Model architecture, such as CNNs, attention mechanisms, elevate category separation, and respond to the distinction between COPD and bronchitis. Feature extraction, such as MFCCs and spectrograms, can capture the acoustic differences of diseases and distinguish asthma from pneumonia. Multimodal fusion and multi-scale analysis alleviate the imbalance and complexity, and the F1 score is above 0.90, which is better than the binary classification of 0.85. Overlapping categories (e.g., COVID-19 vs. flu) remain challenging, and Macro F1 and Cohen's kappa ensure minority class performance, and each class performance needs to be reported to verify reliability. (See chapters III to VI)

Future Directions: Unified evaluation metrics with prioritized reporting on the performance of macros F1, Cohen's kappa, and each category. Develop optimization strategies for a few categories, such as weighted loss functions or oversampling techniques.

D. CROSS-DOMAIN INTEGRATION CHALLENGES

The integration of diverse technical components presents unique challenges in respiratory disease classification. While high-quality datasets combined with advanced feature extraction have improved classification accuracy from 85% to 94% [47], [48], the synergy between different components remains complex. Multi-modal fusion incorporating clinical data enhances diagnostic precision by 7% [30], yet introduces additional computational demands. The relationship between feature complexity and model architecture is particularly evident in graph-based approaches [109], where specialized architectures become necessary for optimal performance. Data augmentation strategies significantly impact both feature quality and model

robustness [206], though balancing feature richness with computational efficiency remains challenging in clinical deployments[207].

Future Directions: Develop lightweight models that support real-time diagnosis with edge computing. Optimize the multimodal fusion algorithm to reduce computing costs and maintain accuracy.

E. MODEL VALIDATION AND REAL-WORLD GENERALIZATION CAPABILITIES

The implementation of AI in healthcare demands robust standardization and quality control to ensure safe and effective clinical integration [208]. Premature deployment of machine learning (ML) models without rigorous external validation and governance can lead to discrepancies between reported and real-world performance, which may ultimately lead to patient harm [209]. To address these risks, defined implementation pathways [210], [211] and a three-stage roadmap for clinical integration of AI have been proposed[212], [213], alongside guidelines that promote standardized reporting [208], [214], [215], [216], [217], reproducibility, and reliability in healthcare AI research. These phases include (1) exploratory model development, (2) a silent trial, and (3) prospective clinical evaluation. Prospective clinical trials, including blinded and multi-center studies, multicenter studies across diverse populations help assess model generalizability [218]. "silent trials" can evaluate AI models prospectively while keeping clinicians blinded, allowing for refinement and addressing issues like dataset drift and stakeholder concerns [208]. Randomized controlled trials (RCTs) remain a gold standard for validating clinical interventions, adhering to principles of randomization, control, and blinding to ensure reliable results [219]. Considering the likelihood of publication bias and the need for more comprehensive research including multicentre trials, diverse outcome measures, and improved reporting standards is crucial [220].

Future Directions: Conduct prospective, blinded clinical trials to evaluate the diagnostic utility of the model in the real world. Perform cross-site validation to ensure the stability of the model in different environments. Design real-world deployment studies to validate the integration of models with clinical workflows.

F. TOWARDS EXPLAINABLE AND TRUSTWORTHY AI

The "black box" nature of deep learning models limits clinical acceptance, making it difficult for doctors to

understand multiple types of prediction bases, affecting trust and regulatory compliance. Interpretive AI tools (such as SHAP, LIME, and saliency maps) can reveal the decision-making process and enhance credibility, but their application in cough analysis is insufficient. Self-supervised learning and federated learning support privacy-preserving training and non-medical audio transfer learning for scalable, non-invasive solutions. Explainable AI (XAI) reveals acoustic signature-driven predictions through SHAP and LIME, such as highlighting high-frequency components of pneumonia, and improving trust in multiple types of diagnostics.

Future Directions: Integration of XAI tools to provide application examples in the field of biomedical acoustics. Develop an interpretive interface for clinicians to simplify the interpretation of model output.

G. RESEARCH-PRACTICE GAP ANALYSIS

Integrating Artificial Intelligence in Clinical Decision Support Systems (CDSS) faces challenges such as workflow integration, data quality, and clinician acceptance [221]. Laboratory models achieving >95% accuracy often prove impractical in resource-constrained clinical settings [222] (As discussed in Section I.C). Real-world environmental factors degrade model performance by 10-15% compared to controlled conditions [59], [61], while privacy requirements substantially restrict data sharing and model optimization. Implementation is further constrained by systemic barriers, including the need to adapt existing healthcare workflows, ensure cost-effectiveness in low-resource environments [223], and manage continuous maintenance demands, all of which present significant challenges. These challenges highlight the critical need for solutions that balance technical sophistication with practical clinical constraints.

Future Directions: Design solutions that adapt to low-resource environments and reduce deployment costs. Provide training for healthcare workers to ensure AI tools are integrated into workflows. Establish a multidisciplinary translational framework to address bottlenecks from algorithm development to clinical deployment.

H. FUTURE RESEARCH DIRECTIONS

In order to advance the diagnosis of multiple types of cough sounds, the following priority areas need to be addressed. As shown in Table VIII.

TABLE VIII
FUTURE RESEARCH DIRECTIONS FOR MULTICLASS COUGH SOUND DIAGNOSIS

Research Domain	Priority Actions
Dataset Development	Standardize recording protocols; include data from children and the elderly; ensure global diversity
Model Optimization	Develop lightweight models; leverage edge computing to support wearable devices
Clinical Validation	Conduct prospective, blinded, multi-center trials; validate in real-world settings
Explainability	Integrate XAI tools (e.g., SHAP, LIME) to enable transparent multiclass predictions
Data Harmonization	Establish standards for crowdsourced and smartphone-based data to enhance reproducibility

Emerging Technologies	Explore self-supervised and federated learning to improve privacy and scalability
Ethics and Regulatory Compliance	Address privacy concerns; ensure compliance with FDA/CE standards for clinical deployment

VIII. CONCLUSION

This systematic review examines 19 significant studies from 2018-2024, and reveals the immense potential of cough sound analysis in multi-class respiratory disease diagnosis. By analyzing datasets, feature engineering, model architectures, and fusion strategies, it establishes a solid foundation for clinical applications. By combining mobile healthcare platforms, wearable devices, quantum computing, and advanced learning techniques, the field has made significant strides towards higher accuracy, greater robustness, and broader applicability.

Future research should focus on Standardizing data collection and annotation protocols, conducting multi-center validation studies, and accelerating the integration of diagnostic systems into existing healthcare workflows for clinical applicability. Additionally, adopting generative modeling and augmentation techniques to balance datasets, developing noise-robust feature extraction methods, and advancing interpretable AI architectures to support transparent clinical decision-making. edge computing technologies and multi-modal fusion strategies will enhance real-time processing and diagnostic accuracy. Privacy-preserving frameworks ensure secure and ethical data handling, addressing concerns related to patient confidentiality and regulatory compliance.

These technologies hold promise for bridging the gap between research and practical implementation, serving as transformative tools in remote and resource-limited settings, and potentially making significant contributions to global respiratory health.

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